

# The Science of Social Networks



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## The Kevin Bacon Game



Boxed version of the  
Kevin Bacon Game

Invented by Albright College students  
in 1994:

- Craig Fass, Brian Turtle, Mike  
Ginelly

Goal: Connect any actor to Kevin  
Bacon, by linking actors who have  
acted in the same movie.

Oracle of Bacon website uses Internet  
Movie Database (IMDB.com) to find  
shortest link between any two  
actors:

<http://oracleofbacon.org/>

## Bacon Number

An Example

Kevin Bacon

Mystic River (2003)

Tim Robbins

Code 46 (2003)

Om Puri

Yuva (2004)

Rani Mukherjee

Black (2005)

Amitabh Bachchan



## Distribution of Actors (According to Bacon Number)

| Bacon Number | Number of Actors | Cumulative Total Number of Actors |
|--------------|------------------|-----------------------------------|
| 0            | 1                | 1                                 |
| 1            | 1,550            | 1,551                             |
| 2            | 121,661          | 123,212                           |
| 3            | 310,365          | 433,517                           |
| 4            | 71,516           | 504,733                           |
| 5            | 5,314            | 510,047                           |
| 6            | 652              | 510,699                           |
| 7            | 90               | 510,789                           |
| 8            | 38               | 510,827                           |
| 9            | 1                | 510,828                           |
| 10           | 1                | 510,829                           |

## The Kevin Bacon Game

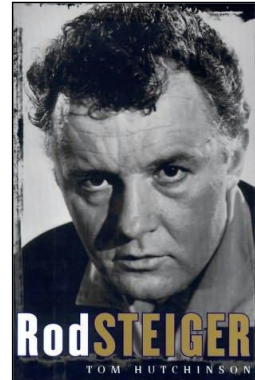
Total # of actors in database:  $\sim 550,000$

Average path length to Kevin: 2.79

Actor closest to “center”: *Rod Steiger* (2.53)

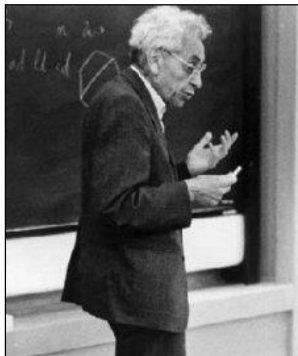
Rank of Kevin, in closeness to center: 876<sup>th</sup>

Most actors are within three links of each other!



Center of Hollywood?

## Erdős Number



Paul Erdős (1913-1996)

Number of links required to connect scholars to Erdős, via co-authorship of papers

Erdős wrote 1500+ papers with 507 co-authors.

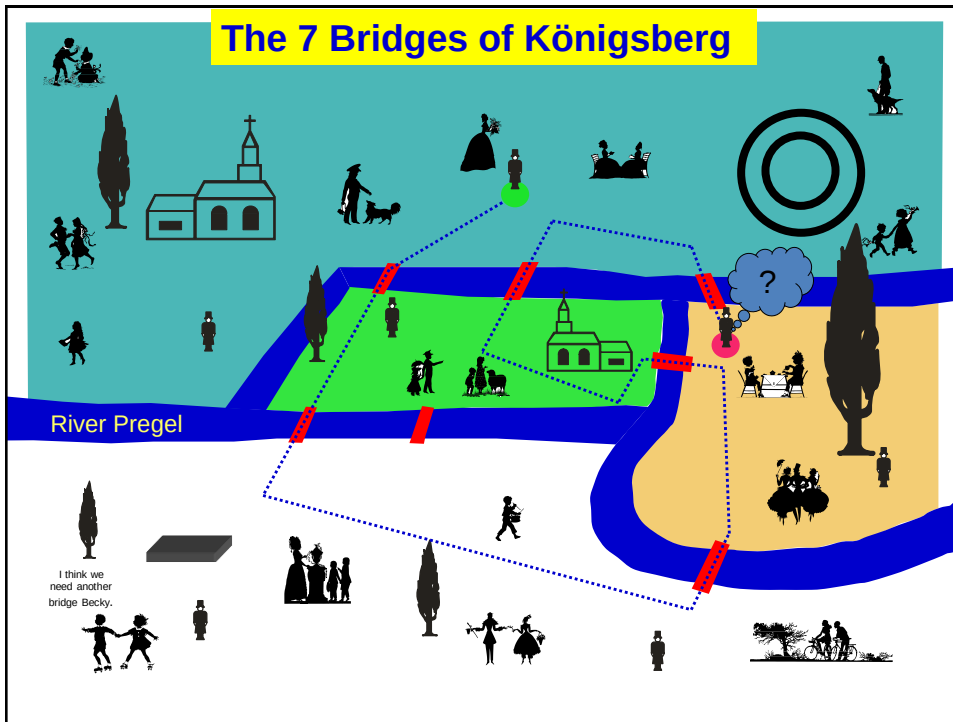
Jerry Grossman's (Oakland Univ.) website allows mathematicians to compute their Erdos numbers:

<http://www.oakland.edu/enp/>

Connecting path lengths, among mathematicians only:

—average is 4.65

—maximum is 13

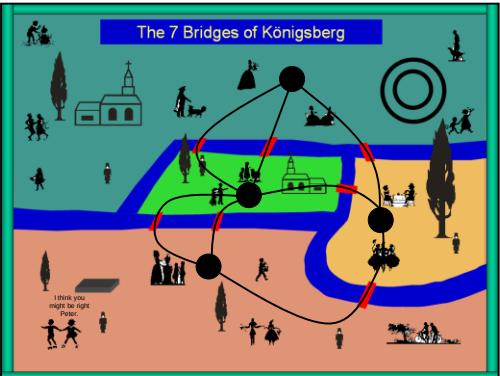




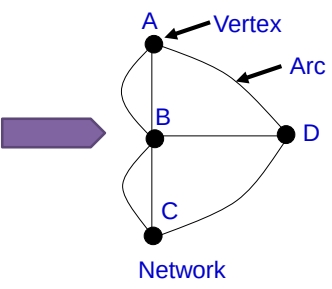
**Euler (1707 – 1783)**

## The 7 Bridges of Königsberg

Although the people of the town felt that such a journey was impossible, they couldn't actually prove it. Euler tackled the problem for them and he decided that he didn't need to make the long journey to Königsberg. Instead, he **abstracted** the problem and made a sketch of the situation.



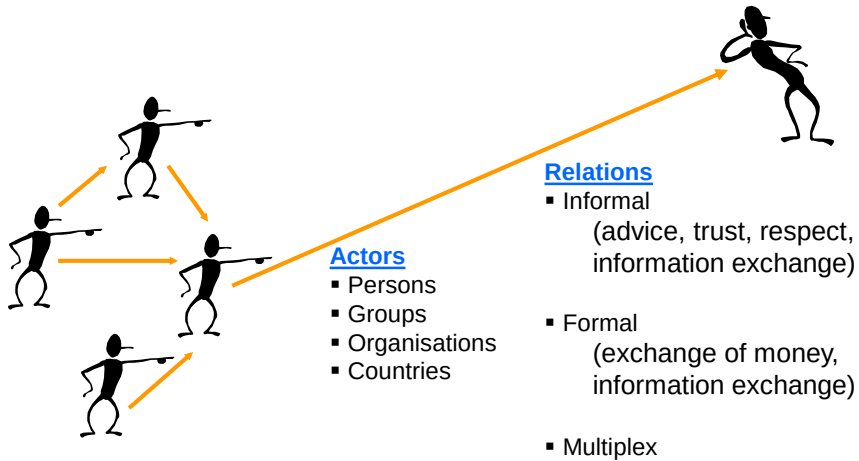
Euler represented the land areas by lettered **vertices** and the bridges by **arcs**. The resulting diagram, he called, a **network**.



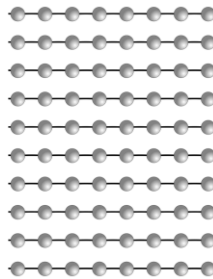
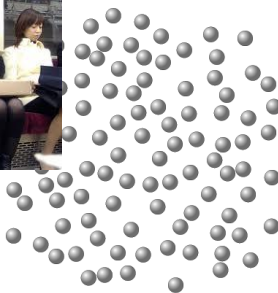
Network

## What are Networks ?

Networks consist of actors connected to one another by relations



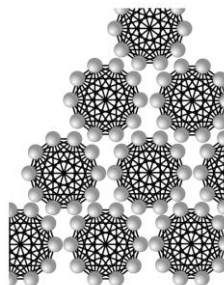
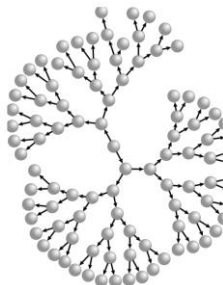
Unconnected Group



Bucket Brigade



Telephone Tree



Military Squads



## Measures of Network Structure

- Network properties
  - density, transitivity
    - measures of complete networks
- Core / periphery
  - categorical, continuous
    - identifies actors in core and those in periphery
- Roles & positions
  - structural equivalence, regular equivalence
    - identifies actors who are similar in their relations with respect to actors in other positions

## Challenges of Networks

### Individual Level

- Strong ties: limited access to new information and knowledge
- "Group thinking"
- Maintaining networks needs efforts

### Organizational Level

- Sharing of roles and responsibilities among partners
- Power structure
- Trust: common rules and practices
- Confidentiality
- Understanding of each others business or activities
- Organizations members of several networks
  - conflicting interest possible
- Maintenance of the network
  - shared vision and strategy
  - trust, trustworthiness
- Cost management

## Strength of Weak Ties

(Granovetter 1973, 1983)

- Actors gain novel information from less intimate ties than close ties
  - actors who are strongly connected share information directly, therefore they possess the same knowledge
  - new information comes from external connections which are likely to be weak

Actors B & D have a weak link - thus are in a more likely position to acquire novel information (which gives them a better opportunity for early adoption)

Tie  
Strength  
Argument

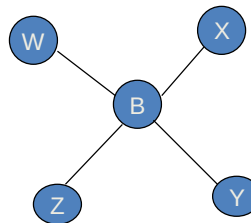
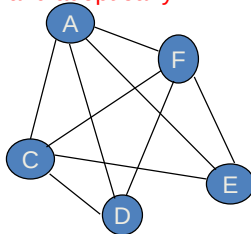


## Structural Holes

(Burt 1992)

- An actor is in a more advantageous position to gain and control novel information if s/he is connected to others who themselves are not directly connected to one another.
  - Nonredundant sources of information allow efficiency with little constraint
  - Developed to explain interpersonal communication in a competitive environment.
- Both actors A & B are connected to 4 others. Actor A is highly constrained with only 1 nonredundant tie (b/n D and E).
- Actor B has 4 nonredundant ties and is in a better position to access novel information and adopt early.

Structural  
Holes  
Argument



## 10 Principles of SNA

1. Networks are Invisible
  - *privacy is protected but marketers are in the dark*
2. People link with others who are Similar
  - *homophily matters*
3. People talk with those who are Physically Close
  - *proximity matters*
4. People who are Similar & Close form Clusters
5. Info quickly Spreads within Dense Clusters
  - *people in the same clique know the same info*
6. Information gets Trapped in Clusters
7. "Bridging Ties" assist Info Flow between Clusters
  - *"opinion leaders" have connections to multiple clusters*
8. Weak Ties are Surprisingly Strong
  - *acquaintances are important sources of novel information*
9. The Net Nurtures Weak Ties
  - *explains why information travels much faster today*
10. Networks go Across Markets
  - *users and non-users talk about multiple product categories*

## Social Network Analysis (SNA)

- Social network analysis is focused on uncovering the patterns of people's interconnectedness and interactions.
- The success or failure of organizations and societies may depend on these patterns.
- Analysis can produce deeper understanding as well as action.
- Applications and current use of SNA is broad and diverse:
  - Knowledge Management
  - Organizational Development
  - Research in economics, social sciences
  - Epidemiology
  - Homeland defense(terrorist networks)

### Some Applications of SNA

- Build a grass root political campaign.
- Identify influential journalists and analysts in the IT industry.
- Map executive's personal network based on email flows.
- Discover the n/w of innovators in a regional economy.
- Realizing 9/11 Al- Qaeda n/w.
- Analyze book selling patterns to position a new book...



Saul Steinberg's  
"View of the World  
from 9<sup>th</sup> Avenue,  
NYC" (1976)

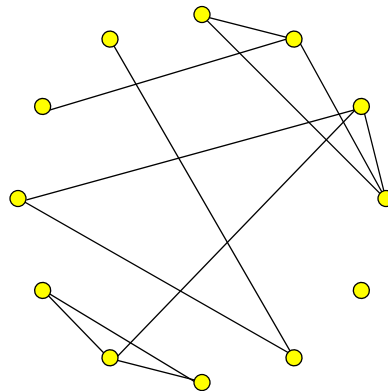


## Society as a Graph

People are represented as  
*nodes*.

Relationships are  
represented as *edges*.

(Relationships may be  
acquaintanceship, friendship,  
co-authorship, etc.)

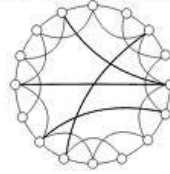


Allows analysis using tools of mathematical graph theory

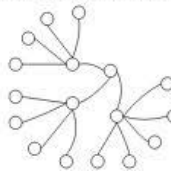
## Different types of networks

- Random
  - Erdos & Renyi
- Small world (six degrees, searchable)
  - Watts, Strogatz, Kleinberg
- Scale free (power law)
  - Albert, Barabasi

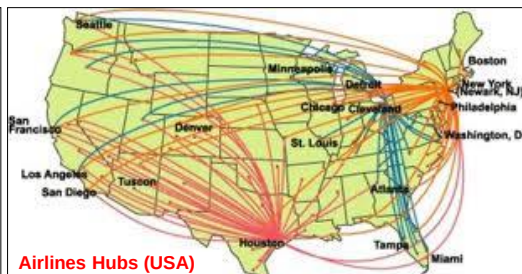
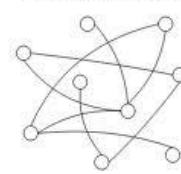
(a) Small-World Network (SWN)



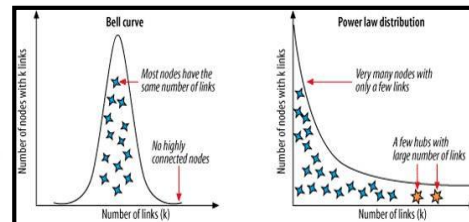
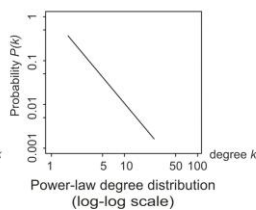
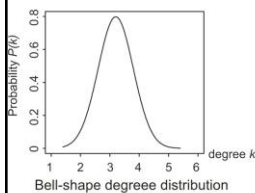
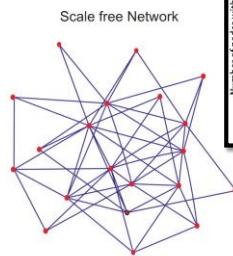
(b) Scale-Free Network (SFN)



(c) Random Network (RN)



## Topological comparison

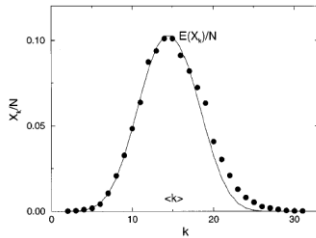


Degree distribution in random networks is bell-shaped.

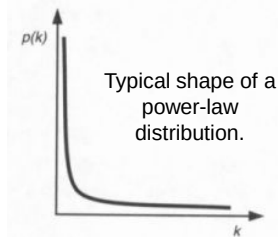
The scale-free network has more high-degree nodes and a power-law degree distribution (which leads to a straight line when plotting the total number of nodes with a particular degree versus that degree in log-log scales).

## Power Laws

(Albert and Barabasi, 1999)



Degree distribution of a random graph,  
 $N = 10,000$   $p = 0.0015$   $k = 15$ .  
 (Curve is a Poisson curve, for comparison.)



Typical shape of a  
 power-law  
 distribution.

What's the degree (number of edges) distribution over a graph, for real-world graphs?

Random-graph model results in Poisson distribution.

But, many real-world networks exhibit a power-law distribution.

## Power Laws



"Map of the Internet" poster

"The rich get richer!"

- Power-law distribution of node distribution arises if
  - Number of nodes grow;
  - Edges are added in proportion to the number of edges a node already has.
- Additional variable fitness coefficient allows for some nodes to grow faster than others.

## Random Graphs

(Erdős and Renyi, 1959)

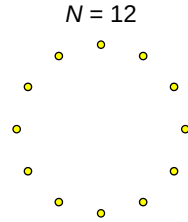
$N$  nodes

A pair of nodes has probability  $p$  of being connected.

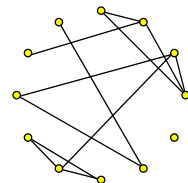
Average degree,  $k \approx pN$

*What interesting things can be said for different values of  $p$  or  $k$ ? (that are true as  $N \rightarrow \infty$ )*

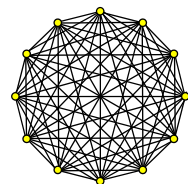
$p = 0.0 ; k = 0$



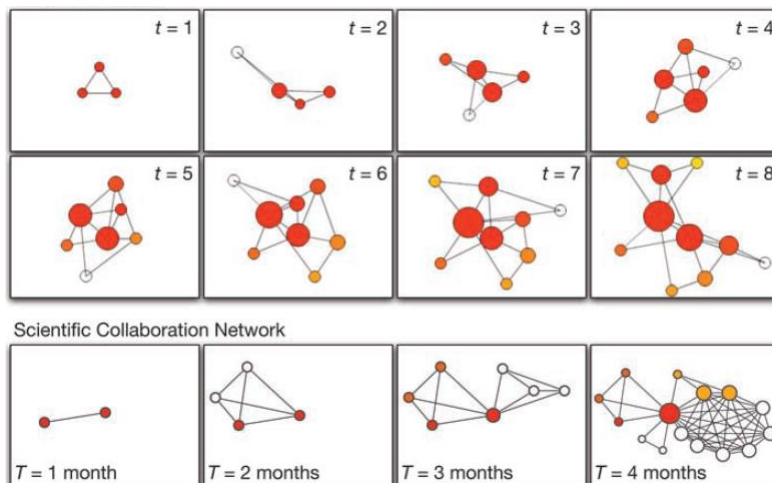
$p = 0.09 ; k = 1$



$p = 1.0 ; k \approx \frac{1}{2}N^2$



## Scale Free Model



*New node precedes to select the node which has more nodes compared with other nodes*

## Six Degrees of Separation (Milgram, 1967)

The experiment:

- Random people from Nebraska were to send a letter (via intermediaries) to a stock broker in Boston.
- Could only send to someone with whom they were on a first-name basis.

Among the letters that found the target, the average number of links was six.



Stanley Milgram (1933-1984)

## Six Degrees of Separation

*“Everybody on this planet is separated by only six other people.  
Six degrees of separation.  
Between us and everybody else on this planet.  
The president of the United States. A gondolier in Venice...  
It's not just the big names. It's anyone.  
A native in a rain forest. A Tierra del Fuegan. An Eskimo.  
I am bound to everyone on this planet by a trail of six people...”*



***Six Degrees of Separation,  
a play by John Guare***



## Three Degrees of Influence

The influence of actions ripples through networks 3 degrees  
(to and from your friends' friends' friends)

*You are affected by people you don't even know!*

**Influence dissipates after 3 degrees because:**

- ❑ **Intrinsic decay:** corruption of information (*like the game telephone*)
- ❑ **Network Instability:** social ties become unstable at 4+ degrees of separation
- ❑ **Evolutionary Purpose:** we evolved in small groups where everyone was connected by 3 degrees or less

## If you wannabe Happy!



- A person is 15% more likely to be happy if directly connected to a happy person (1 degree)
- At 2 degrees 10% more likely to be happy
- At 3 degrees 6% more likely to be happy
- Each unhappy friend decreases the likelihood of happiness 7%

An increase of \$10,000 of income per year yields only a **2%** increase chance in happiness. Compare that to a **15%** chance from a happy friend and a **6-10%** from someone you may have never met, but to whom you are indirectly tied!

Long term happiness is affected by:

- 50% genes
- 10% circumstance (i.e. quality of life)
- 40% attitude (what you think and do)

**Happy families are all alike;  
every unhappy family is unhappy  
in its own way.**

**-Leo Tolstoy,  
"Anna Karenina" (1877)**





## Relative Standing Phenomenon!

Experiment in the workplace:

Job 1: Earn a \$33,000 salary with everyone else getting \$30,000

Job 2: Earn \$35,000 and have fellow colleagues earn \$38,000

- **Majority preferred Job 1 (Big Fish in a small pond)**

Another experiment dealing with attractiveness - participants were asked which state was more desirable:

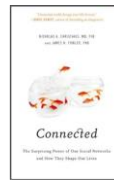
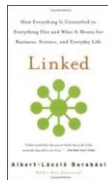
A: Your physical attractiveness is 6; others average 4.

B: Your physical attractiveness is 8; others average 10.

- **75% of people preferred A**
- **93% of Harvard Grads preferred A**



## Credits



- Albert, Reka and A.-L. Barabasi. "Statistical mechanics of complex networks." *Reviews of Modern Physics*, vol.74, no.1, pp.47-94, 2002.
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- **Watts, Duncan, *Six Degrees: The Science of a Connected Age*, W.W.Norton, 2003.**